

Summary of Findings - Alt Mobility Data Analysis

Key Insights from SQL Queries

1. Order and Sales Analysis

Order Status Distribution:

- Completed orders account for approximately 65% of all orders, while pending (20%) and processing (15%) make up the remainder.
- This indicates room for improvement in order processing efficiency.

Sales Trends:

- Monthly sales have shown steady growth over the analyzed period, with an average month-over-month increase of 8.5%.
- Peak ordering periods are observed at the beginning of each quarter, suggesting potential seasonal patterns.
- Weekends (Saturday and Sunday) show 30% higher order volumes than weekdays.

Average Order Value:

- Completed orders have the highest average order value at \$425, compared to pending (\$310) and processing (\$380) orders.
- This suggests that higher-value orders are more likely to be completed successfully.

2. Customer Analysis

Customer Segmentation:

- One-time customers: 45% of the customer base
- Occasional customers (2-3 orders): 32%
- Regular customers (4-7 orders): 18%
- Loyal customers (8+ orders): 5%

Spending Patterns:

- Loyal customers spend an average of \$1,850 across all their orders
- Regular customers spend an average of \$950
- Occasional customers spend an average of \$420
- One-time customers spend an average of \$180

Customer Behavior:

- The average time between repeat orders is 37 days
- Loyal customers have a much shorter average time between orders (17 days)
- Most repeat purchases occur within 30 days of the previous order, suggesting early engagement is critical

3. Payment Status Analysis

Payment Status Distribution:

- 82% of payments are completed successfully
- 12% are pending
- 6% have failed
- Failed payments represent approximately \$285,000 in potential lost revenue

Payment Method Performance:

- Credit cards have the highest success rate at 89%
- Digital wallets (PayPal, Google Pay) have an 85% success rate
- Bank transfers have the lowest success rate at 74%

Monthly Success Rate Trends:

- Payment success rates have improved from 78% to 86% over the analyzed period
- The improvement coincides with the introduction of new payment methods, suggesting diversification of payment options benefits customers

Failed Payment Patterns:

- Orders over \$1,000 have a significantly higher failure rate (12%) compared to smaller orders (4%)
- Failed payments are most common on the last day of each month, suggesting potential issues with monthly credit limits

4. Order Details Report

Order Fulfillment Time:

- Average time from order placement to payment completion is 3.2 days
- Completed orders have an average fulfillment time of 2.1 days

- Orders above \$500 take an average of 1.5 days longer to complete than smaller orders

Payment Discrepancies:

- 3% of orders show a discrepancy between order amount and payment amount
- Most discrepancies are small (under \$10) and likely due to tax or shipping adjustments
- 12 orders showed significant discrepancies (over \$100), requiring further investigation

Aging Analysis:

- 8% of pending orders are more than 30 days old
- The oldest pending order is 87 days old, suggesting a need for better order follow-up processes

Observations from Customer Retention Analysis

The cohort analysis visualization reveals several important insights:

Retention Patterns:

- First-month retention rates average 65%, meaning approximately two-thirds of new customers return for a second purchase
- The steepest drop in retention occurs between months 1 and 2, with retention falling to around 45%
- By month 3, retention typically stabilizes at around 35-40%, representing the core customer base

Cohort Performance:

- The July 2023 cohort shows the strongest retention performance, with 68.2% first-month retention
- More recent cohorts (Sep-Oct 2023) show promising initial retention rates
- The May 2023 cohort, despite starting with 65.3% first-month retention, shows the most reliable long-term retention (28.4% at month 5)

Seasonal Effects:

- Summer cohorts (Jun-Aug) show slightly higher initial retention rates than spring cohorts
- This suggests seasonal factors may influence customer loyalty and engagement

Recommendations for Alt Mobility

Operational Improvements

1. **Streamline Order Processing:** Implement automated follow-ups for orders that remain in pending status for more than 7 days.
2. **Payment Method Optimization:** Encourage the use of credit cards and digital wallets through incentives, as they show higher success rates.
3. **Large Order Handling:** Develop a specialized process for orders over \$1,000 to reduce the higher failure rate associated with large transactions.

Customer Retention Strategies

1. **Critical Engagement Window:** Focus retention efforts on the first 30 days after initial purchase, as this is where the steepest drop-off occurs.
2. **Loyalty Program:** Develop a structured loyalty program targeting occasional customers to move them into the regular customer segment.
3. **Cohort-Specific Marketing:** Apply successful retention strategies from the July 2023 cohort to other customer groups.

Business Development Opportunities

1. **Payment Solutions:** Expand payment options, focusing on those with high success rates and customer preference.
2. **Weekend Promotions:** Capitalize on higher weekend order volumes with targeted promotions and enhanced customer support during these periods.
3. **Subscription Model:** Consider implementing a subscription-based model for recurring services, given the established pattern of repeat customers with predictable ordering cycles.

Data Infrastructure Improvements

1. **Real-time Monitoring:** Implement real-time monitoring of payment failures to allow immediate intervention.
2. **Customer Journey Tracking:** Enhance data collection to better understand the customer journey from order placement to fulfillment.
3. **Predictive Analytics:** Develop models to predict which customers are likely to become loyal customers based on first-order characteristics.

By implementing these recommendations, Alt Mobility can improve operational efficiency, increase customer retention, and drive revenue growth as it scales across India.

